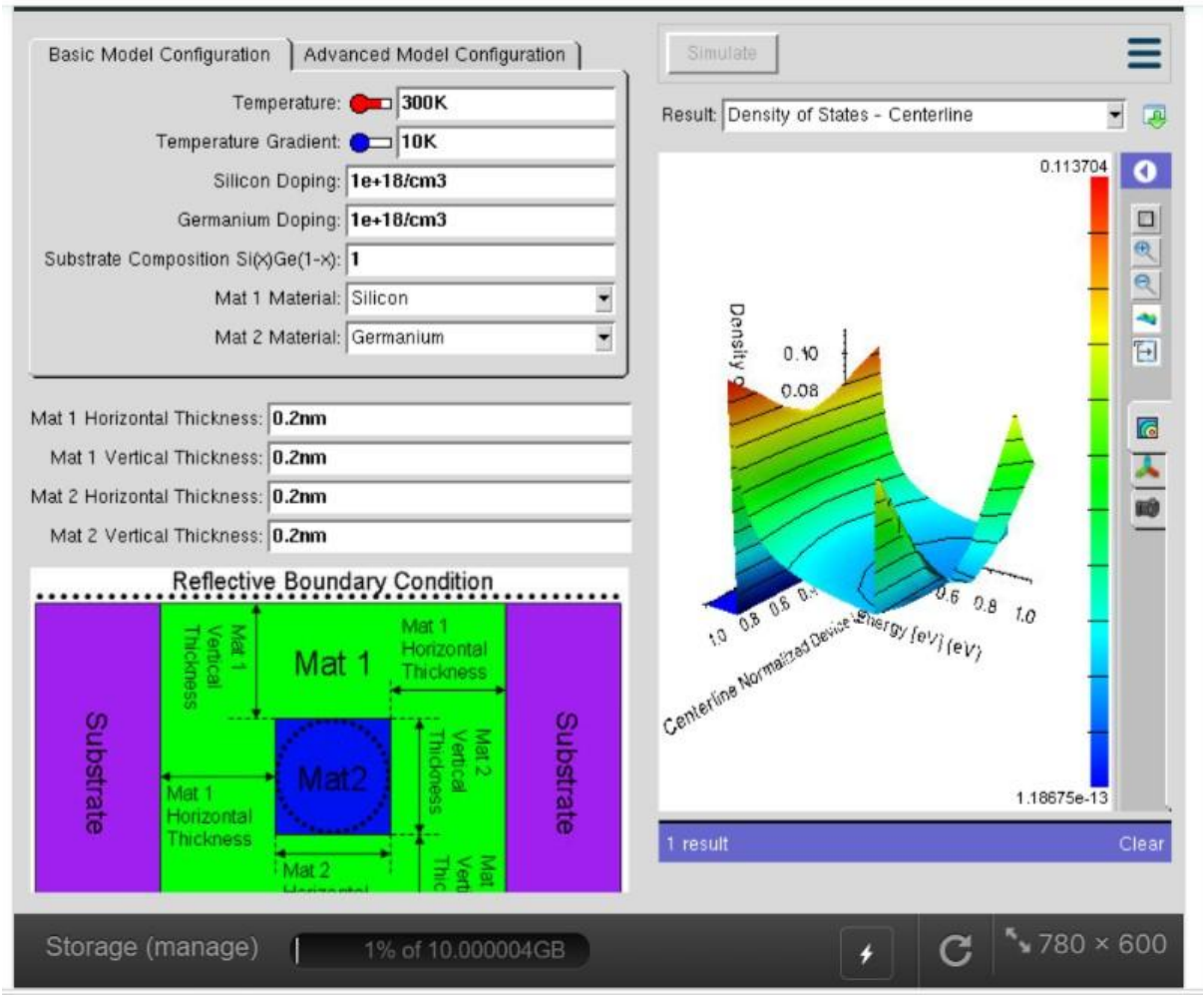
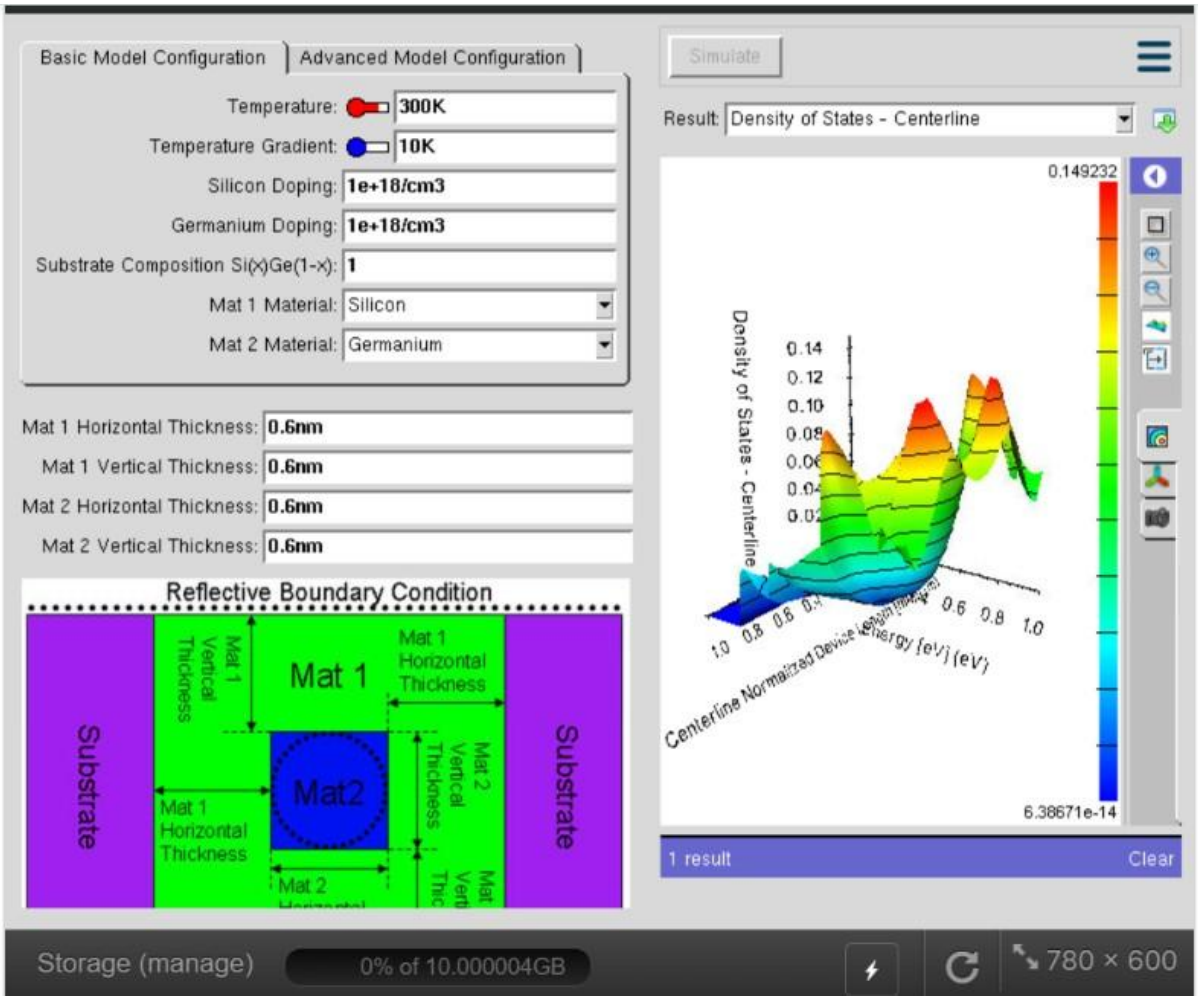
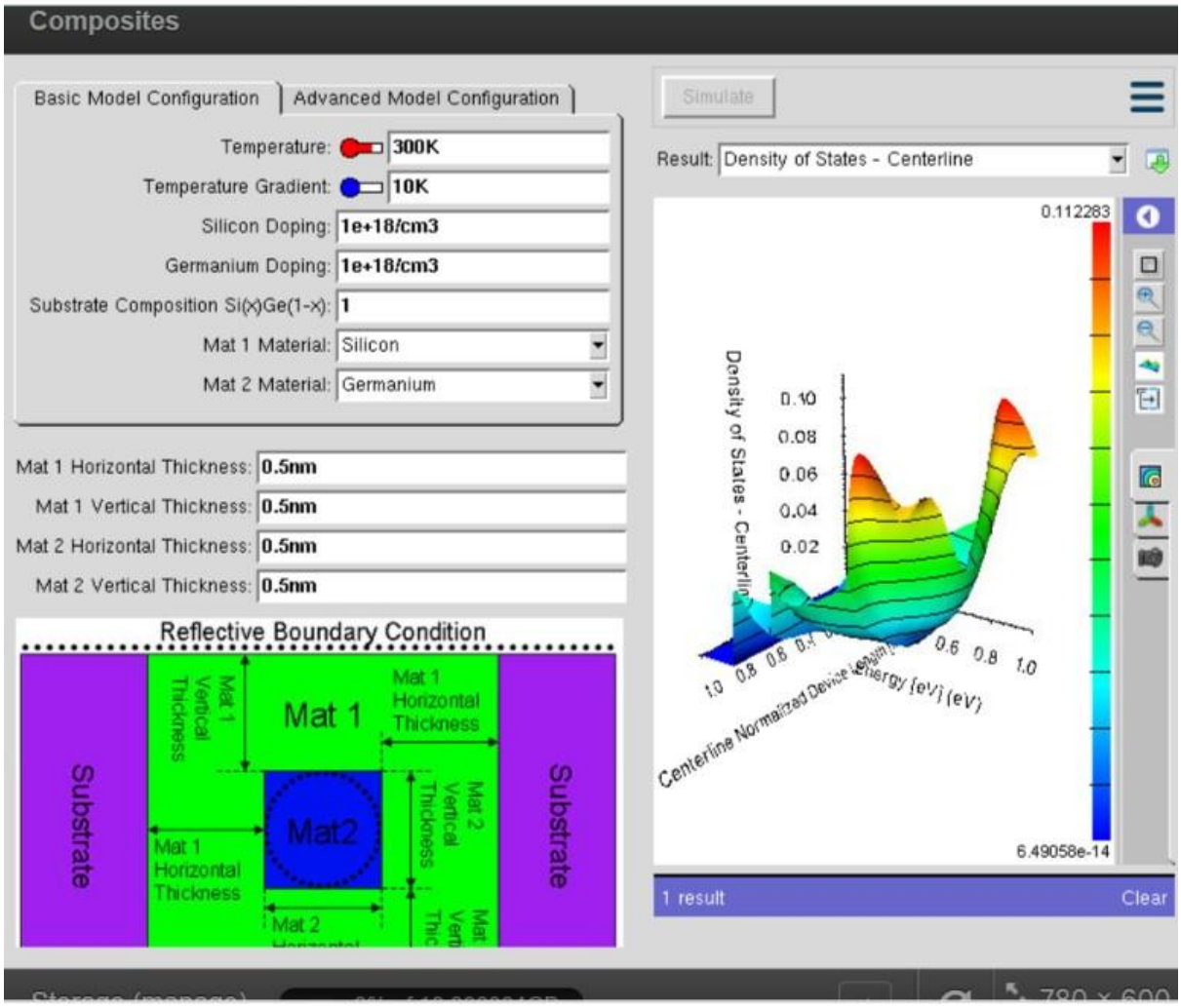
EXPERIMENT 3 SIMULATION OF DENSITY OF STATES OF QUANTUM WELL STRUCTURE USING NANOHUB




Basic Model Configuration
Advanced Model Configuration
Temperature: 300K
Temperature Gradient: 10K
Silicon Doping: 1e+18/cm3
Germanium Doping: 1e+18/cm3
Substrate Composition Si(x)Ge(1-x): 1
Mat 1 Material: Silicon
Mat 2 Material: Germanium
Mat 1 Horizontal Thickness: 0.6nm
Mat 1 Vertical Thickness: 0.6nm
Mat 2 Horizontal Thickness: 0.6nm
Mat 2 Vertical Thickness: 0.6nm
Reflective Boundary Condition
Substrate
Mat 1
Mat2
Substrate
Simulate
Result: Density of States - Centerline
0.149232
6.38671e-14
Density of States - Centerline
Centerline Normalized Device Length [nm]
Energy [eV] (eV)
1 result
Clear
Storage (manage)
0% of 10.000004GB
780 × 600


Composites
Basic Model Configuration
Advanced Model Configuration
Temperature: 300K
Temperature Gradient: 10K
Silicon Doping: 1e+18/cm3
Germanium Doping: 1e+18/cm3
Substrate Composition Si(x)Ge(1-x): 1
Mat 1 Material: Silicon
Mat 2 Material: Germanium
Mat 1 Horizontal Thickness: 0.5nm
Mat 1 Vertical Thickness: 0.5nm
Mat 2 Horizontal Thickness: 0.5nm
Mat 2 Vertical Thickness: 0.5nm
Reflective Boundary Condition
Substrate
Mat 1
Mat2
Substrate
Mat 1 Vertical Thickness
Mat 1 Horizontal Thickness
Mat 2 Vertical Thickness
Mat 1 Horizontal Thickness
Mat 2
Simulate
Result: Density of States - Centerline
0.112283
Density of States - Centerline
0.10
0.08
0.06
0.04
0.02
Centerline Normalized Device Length
Energy (eV) (eV)
6.49058e-14
1 result
Clear